Christian Scherer

Full-Stack Systems Architect | Cloud Infrastructure, Zero Trust Networking, Automation Design, Systems Integration & Software Development
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Currently living in Peculiar, Missouri, USA — relocating to Brighton / Greater London, UK
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PROFILE

Full-stack and AI systems engineer building software, automation, and AI tooling that runs enterprises. Over eight years at Burns & McDonnell modernizing Microsoft 365 and ServiceNow at enterprise scale, now bridging IBM i platforms to Azure at Murphy-Hoffman Company. n8n and LLM pipelines, RouteLLM routing, OpenCode/DeepSeek agentic development, multi-agent systems, and local LLMs — all deployed governance-first under Zero Trust, with disposable-container verification gates and E2E-encrypted remote tooling. Strong across Python, TypeScript, .NET, React, FastAPI, Docker, PostgreSQL, and CI/CD.

SKILLS

Full-Stack Development — Advanced: Python, TypeScript, React, Next.js, .NET, C#, FastAPI, SQL, PostgreSQL, Vite, Vercel
AI / LLM Engineering — Advanced: LLMs, RAG / retrieval, Prompt engineering, Multi-agent systems, RouteLLM, OpenAI-compatible API, Ollama, OpenCode, DeepSeek, Agentic development
Automation Orchestration — Expert: n8n, NocoDB, JobSpy, PowerShell, ServiceNow, Workflow Automation, pytest, Browserless
Cloud & Infrastructure — Advanced: Azure, Docker, Linux, Cloudflare Zero Trust, Terraform, CI/CD, GitHub Actions, Neon Postgres, Render, Vercel
Enterprise Platforms — Expert: Microsoft 365, Exchange Online, Entra ID, Intune, Active Directory, SCCM
Legacy & Integration — Advanced: IBM i, RPG/CL, System i, REST APIs, OAuth, Azure tokens

EXPERIENCE

Programmer/Analyst — Murphy-Hoffman Company (MHC Kenworth) (2025-10–Present)
Developed and enhanced business applications on the IBM i platform using RPG and CL programming; integrated modern technologies like Azure, Python, and .NET with legacy systems to boost efficiency; collaborated with internal stakeholders to design and implement solutions that streamline operations.
Staff Software Developer — Burns & McDonnell (2023-03–2025-09)
Designed, coded, and tested software for business challenges; led architecture and modernization incl. QTS modernization for Azure hosting; built secure REST API integrations (OAuth, Azure tokens); automated IT processes (licensing, storage archiving, data normalization) saving hundreds of admin hours annually; mentored developers via code reviews.
Staff IT System Specialist — Burns & McDonnell (2023-01–2023-03)
Directed ServiceNow architectural changes: new catalog modules and automation streamlining request handling for thousands of users; integrated workflows with Azure-hosted services incl. ESRI backup automation; documented technical decisions with governance compliance.
Assistant IT System Specialist — Burns & McDonnell (2022-07–2023-01)
Developed and implemented new application modules and service catalog workflows within the Now platform, enhancing system efficiency; analyzed business requirements and documented technical solutions to improve system functionality; troubleshot technical issues throughout the Software Development Life Cycle, ensuring smooth operation.
Assistant IT Applications Specialist — Burns & McDonnell (2021-08–2022-07)
Troubleshot critical issues and assisted with problem resolution for customers; developed and maintained knowledge base articles to enhance user support; researched, verified, and documented product defects to improve overall product quality.
End User Support Technician — Burns & McDonnell (2017-06–2021-08)

Deployed computer software systems to engineers locally and in the field; assisted engineers with software and hardware issues occurring with their computer systems.

Laptop Repair Technician — Heartland Services Inc (2016-08–2017-06)

Diagnosed and repaired computers and tablets for Panasonic, including hardware and software fixes; conducted board and LCD replacements and other hardware fixes to ensure optimal device performance; collaborated with team members to streamline repair processes and improve efficiency in diagnosing issues.

PROJECTS

Secure-Term — Encrypted Browser Terminal

E2E-encrypted long-poll browser terminal that keeps terminal content opaque to edge and TLS-inspection proxies: Python stdlib HTTP server, AES-GCM (WebCrypto), paramiko SSH into persistent tmux sessions, constant-time auth, no inbound ports.

Agent-Test — Disposable-Container Verification Pipeline

Self-contained CI harness that runs static analysis, unit/API/UI tests, and hardcoded-value scans in ephemeral containers without touching live services; gates merges on reproducible results.

Job Search Automation Dashboard — https://jobdashboard.christianscherer.dev

Automated job-search pipeline: daily JobSpy scraping, a verified evidence-backed skill inventory, LLM-based fit scoring, tailored resume PDF + cover letter generation, and company research with per-job study guides and interview prep (n8n, NocoDB, FastAPI, Python).

RouteLLM CLI Development Pipeline — https://routellm.abacus.ai

Integrated RouteLLM, an OpenAI-compatible LLM routing gateway, as the single endpoint in the dev pipeline for agentic and automation workflows.

Local LLM / AI Agent Orchestration

Homelab stack orchestrating local LLMs and multi-agent workflows with Ollama; containerized, isolated, accessed via Zero Trust.

Zero Trust Homelab Infrastructure

Security-hardened homelab: Cloudflare Zero Trust access, containerized services, automated patching, secure remote access.

IBM i to Azure Integration Bridges

Bridging core IBM i (RPG/CL) business systems to Azure with Python/.NET integration layers, modernizing legacy platforms without rewriting.

Personal Portfolio Platform — https://christianscherer.dev

React/TypeScript portfolio site with an ASP.NET Core API and Postgres backend, including an admin portal for project management.

Track-Folio

Full-stack application-tracking solution web app.

Project Management API & Database

ASP.NET Core Web API with Postgres serving the portfolio platform.

M365 Tenant Automation & Governance

Modernize and automate a Microsoft 365 tenant: automation and governance tooling at enterprise scale.

Active Directory Data Pipeline Automation

Automation pipeline for Active Directory data extraction, normalization, and reporting.

EDUCATION

Southern New Hampshire University — Computer Science (2020–2022)

Coastal Carolina Community College — Computer Programming (2013–2015)

University of North Carolina at Charlotte — Computer Science (2012–2013)